ECON 3420A Assignment 1: Build Your Own Portfolio

Lecture Note Part 1-3 | 100 marks

Question Structure: There are 6 question in total. Each question includes two sections:

- **Task to AI** requires you to direct AI agent to write codes and generate outputs (tables, figures, etc), etc;
- **Task to Student** requires you to make preparations for working with AI, review and explain outputs with financial economics knowledge.

Submit Only Three Files:

1. Your Completed **`portfolio_model.py`**. When you direct AI to work on tasks, you should **complete 5 Python functions** in **`portfolio_model.py`** (you could see the corresponding question numbers):
   - `estimate_index_model` (Q2),
   - `single_index_covariance` (Q3),
   - `optimize_risky_portfolio` (Q4),
   - `infer_risk_aversion` (Q5),
   - `allocate_portfolio` (Q6).
2. Your **Final QMD and PDF Reports**. Copy the template file `solution_template.qmd` and rename it with `studentID_assignment_one.qmd`. You should include answers for all question in the `qmd` file, and call each function above under corresponding question answering area. Render the `qmd` file into a PDF file with same file name. See more details in `Quarto and QMD.pdf`.

Note:

- **Software/Packages Check** Before You Start: **Quarto, Python, and packages in `requirements.txt`**. See `data_guide.pdf` for installation, data download and the Question 5 quiz.
- The quality of your interactions with AI and explanations of outputs will be graded, but not the AI chat length. You may let AI tutor you or polish your final PDF report.

Before you answer Q1-Q6

Each item below is part of your answer to the question in brackets. `solution_template.qmd` has a place for each.

- **(a) Beta prediction (Q2): before** running any regression, record which of your five stocks you expect to have the highest beta, your reason, and the date and time. After Question 2, record what you observed.
- **(b) AI evidence entry (Q6):** for the half-premium test in Question 6, fill in a four-row table: the AI tool you used and its relevant claim (quoted); your prediction before the test; the fixed inputs and the original and new risky shares; and whether you accept or revise the AI's recommendation.

- **(c) Data sources (Q1):** in a few lines, state your data sources, five tickers and benchmark, currency, sample dates, price adjustments and bill-rate convention. Say if you used the supplied practice data.

1. Choose stocks and calculate returns (15 marks)

Task to Student:

- Choose five company stocks in one market/currency and a broad market index or tracking fund (ETF) in the same currency. Give names, tickers, sectors and your reasons. The benchmark is a comparison, not a sixth holding.

Task to AI:

- Obtain online at least **60 consecutive monthly returns on matching dates**. The example horizon is October 2021-September 2026, requiring data for the day before October 2021 as starting point.
- Calculate returns and show average monthly return and volatility (standard deviation) in a table. **Excess return** means return minus the safe return. Use the previous month's last bill yield.

2. Estimate and understand alpha and beta (20 marks)

Task to AI: estimate the single-index model from Part 3 for each stock by OLS, with an intercept:

$$X_{i,t} = \alpha_i + \beta_i X_{M,t} + e_{i,t}.$$

Here $X_{i,t} = R_{i,t} - R_{f,t}$ and $X_{M,t} = R_{M,t} - R_{f,t}$ are stock and market excess returns; $e_{i,t}$ is the unexplained part. Prepare a table of $\hat{\alpha}_i$, $\hat{\beta}_i$ and R^2, and one chosen stock's graph with its fitted security characteristic line (SCL). Put $X_{M,t}$ horizontally and $X_{i,t}$ vertically. Alpha is the intercept, beta is the slope; hats denote estimates, and R^2 measures how much variation the line explains.

Task to Student: explain the chosen stock's $\hat{\alpha}_i$, $\hat{\beta}_i$ and R^2. If market excess return rises by one percentage point, what change does the model predict? Why does the line not predict every monthly return? Was your initial prediction supported?

3. Understand variance and covariance (20 marks)

Variance measures how much returns vary; covariance describes how two stocks move together. A **residual** is the actual excess return minus the regression's fitted value. This model assumes different stocks' residuals do not move together (zero covariance).

Task to AI: use the variance and covariance formulas from Part 3:

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma^2(e_i).$$

For two different stocks, $i \neq j$:

$$\mathbf{Cov}(X_i, X_j) = \beta_i \beta_j \sigma_M^2.$$

Here σ_M^2 is market variance and $\sigma^2(e_i)$ is residual variance. Use estimated betas, the market's sample variance, and the residual estimate from Part 3:

$$\hat{\sigma}^2(e_i) = \frac{\sum_{t=1}^{T} \hat{e}_{i,t}^2}{T-2},$$

where T is the number of months. Build **one five-by-five table**: variances on the diagonal, covariances elsewhere, with ticker labels.

Task to Student: choose one covariance entry and explain its sign. Why can combining stocks that move differently reduce risk? Give one reason why the assumption about residuals might fail for your stocks. No proof is required.

4. Find and assess a risky portfolio (20 marks)

The Sharpe ratio measures extra return above the safe asset per unit of risk:

$$\mathrm{SR}_P = \frac{\mathbb{E}(R_P) - R_f}{\sigma_P}.$$

Here P is the risky portfolio, R_f the safe rate and σ_P its volatility. The numerator can be estimated from average excess returns.

Task to AI:

- Use historical average excess returns and the risk table from Question 3. For annual figures, multiply monthly means and each table entry by **12** (thus volatility, as squared root of variance, increases by $\sqrt{12}$).
- Find the five stock weights giving the **highest Sharpe ratio**. Weights must be nonnegative and total **100%**; zero weights are allowed.
- Report weights, annual excess return, volatility and Sharpe ratio. Compare with putting 20% in each stock.

Task to Student: Check whether the Sharpe ratio of the portfolio formed with the weights found above is at least as high as that of the equally weighted portfolio. Explain what you notice about the holdings and why these historical results cannot guarantee future performance. You do not need to derive the optimization

5. Work out your risk aversion (10 marks)

Use **your Question 4 portfolio** P. Let $\delta = \mu_P - R_f$ be its estimated annual excess return, σ_P^2 its annual variance. Use the latest saved annual bill rate R_f and record its date.

A **certainty equivalent** is the sure return c you value equally to P, with formula $c = \mu_P - \frac{\gamma}{2}\sigma_P^2$. For $\delta > 0$, define $G = \delta/\sigma_P^2$ and corresponding risk aversion level $\gamma_m = mG$. In this question, AI will construct certainty equivalent rates c_m and wealth levels under different levels of γ_m (suppose initial wealth level is 10,000) with the formula:

$$c_m = \mu_P - \frac{\gamma_m \sigma_P^2}{2}, \qquad W_m = 10000(1 + c_m).$$

and you, as portfolio manager, are required to select target certainty equivalent wealth W_m. Then AI could derive implied risk aversion level γ_m for you.

Task to AI: save P's estimates and weights, then run `risk_aversion_quiz.py` (ensure that it imports P's statistics correctly). Show P's mean and volatility, the lowest 10%, middle 80% and highest 10% return/wealth ranges, and the certainty equivalent wealth W_m for five risk-aversion levels γ_m, labelled with letters A to E. The python script will input the student's letter selection

whose certainty equivalent wealth W_m meets the student's target, report the implied risk aversion γ_m as student's risk aversion, and save the choice and portfolio inputs.

Task to Student: read the certainty equivalent menus printed by `risk_aversion_quiz.py` and select the **certainty equivalent wealth closest to your target**. Explain your choice and what a larger gamma means. This is a **coarse estimate from a restricted menu**; say if the options fit your preferences poorly. If $\delta \leq 0$, explain the exception in Question 6 instead of supplying a numeric gamma.

6. Choose your stock-and-safe-asset mix (15 marks)

In this question, you construct a **final portfolio** F that combines safe asset and the Question 4 risky portfolio P. Using **the same portfolio weights, excess return** δ and variance σ_P^2 from Question 4, and your implied gamma from Question 5, the optimal share of wealth in P is

$$w_r^* = \frac{\delta}{\gamma \sigma_P^2},$$

and the remaining $1 - w_r^*$ is held in the safe asset.

Task to AI:

- Calculate w_r^* directly. For wealth of 10,000, tabulate F: the percentage $w_r^* w_i$ and amount $10000\, w_r^* w_i$ held in each stock i, and the percentage $1 - w_r^*$ and amount held in the safe asset.
- Report F's annual expected return $\mathbb{E}(R_F) = R_f + w_r^* \delta$ and volatility $\sigma_F = w_r^* \sigma_P$, using the same dated R_f as Question 5.
- After the student records a prediction, halve P's **expected excess return** δ, keeping stock weights, variance and gamma fixed, and recalculate the risky share.
- **Nonpositive-premium exception:** if $\delta \leq 0$, positive risk aversion cannot give an interior optimum. Keep the estimate and explain it, skip numeric gamma elicitation, and report 100% in the safe asset. The half-premium test also gives a zero risky share. Do not replace the estimate to force an interior result.

Task to Student: explain how risk aversion affects the risky share. Predict the half-premium's effect **before** AI runs it, then compare the results. In at most 100 words, explain whether you would accept the allocation based on these estimates.